

1 Digital Design

- nFET = Not Negated. Turns on by 1 or High.
- pFET = Negated. Turns on by 0 or Low.
- Xor = $\bar{A}B + A\bar{B}$
- Fan-out refers to the number of additional gates that can be physically connected to the output of a gate.
 - I_{ILMax} = The max current draw of a gate input for a low input value
 - I_{OLMax} = The max current output of a gate for a low output
 - I_{IHMax} = The max current draw of a gate input for a high input value
 - I_{OHMax} = The max current output of a gate for a high output
 - Max Fanout = $Min(\frac{I_{OLMax}}{I_{ILMax}}, \frac{I_{OHMax}}{I_{IHMax}})$
- We don't do POS for Variable Entered K-Maps. Turn into SOP.
- Inside Each Comparator
 - $out_gt = in_gt + (in_eq * a * \bar{b})$
 - $out_lt = in_lt + (in_eq * \bar{a} * b)$
 - $out_eq = in_eq * (a \oplus b)$